

# CAMPUS MATE – A COLLEGE AI

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**Abstract:** *The Campus Mate – A College AI is a web-based chatbot developed to help students and visitors easily find locations within a college campus. Large campuses often make it difficult for new students and guests to locate departments, laboratories, offices, or other facilities. This project provides an interactive chatbot interface where users can ask questions about the location of various places in the college. The system is developed using the Django web framework, along with HTML, CSS, and JavaScript for the front-end interface. Users can type their queries in the chat window, and the bot responds with the appropriate location information. Additionally, a map feature is integrated that allows users to view the location visually by clicking a map icon displayed next to the bot's response. The main objective of this project is to improve accessibility and reduce confusion for students, staff, and visitors by providing quick and easy directions within the campus. The chatbot interface makes the system user-friendly and interactive, enabling users to obtain information instantly without needing manual assistance. The main objective of this project is to reduce confusion, save time, and provide quick assistance to users navigating the campus. The application also includes user authentication features such as login and logout to ensure secure access to the system. The chat interface is designed to be simple, responsive, and user-friendly so that users can easily interact with the bot without technical knowledge. The integration of mapping functionality helps users understand directions more clearly compared to text-based instructions alone. By combining chatbot technology with web development and map integration, the system acts as a virtual campus guide that can provide instant responses to location queries. The chatbot processes the query and returns the relevant information instantly. This system reduce the workload of college staff while ensuring reliable access to the students. This project presents the design and development of a web-based College Location Chatbot that assists users in identifying and locating facilities within a campus environment. This project demonstrates how intelligent chatbot systems can improve user experience in educational institutions by offering automated guidance and improving accessibility to campus information.*

**Keywords:** Django Framework, Web Application, Chatbot System, Campus Navigation, College Location Assistance, User Query Processing, Google Maps Integration, Interactive Chat Interface, Location-Based Services, Automated Response System, Web Development, User-Friendly Interface, Data Handling, Real-Time Query Response.

## I. INTRODUCTION

In large educational institutions, navigating the campus and locating specific departments or facilities can often be difficult for new students, visitors, and sometimes even staff members. Colleges usually have multiple buildings, blocks, laboratories, administrative offices, and other facilities spread across a wide area. As a result, people frequently require assistance to find the correct location within the campus. To address this issue, the College Location Bot has been developed as a web-based chatbot system that helps users easily find different locations inside the college. The chatbot allows users to ask questions related to campus locations through a simple chat interface. Based on the user's query, the system provides appropriate responses that guide the user to the requested place.

The project is developed using the Django web framework, which is widely used for building secure and scalable web applications. The front-end of the system is designed using HTML and CSS, creating an interactive and user-friendly chat interface. This enables users to communicate with the chatbot in a natural and convenient way. The project is developed using the Django web framework, which is widely used for building secure and scalable web applications. The front-end of the system is designed using HTML and CSS, creating an interactive and user-friendly chat interface. This enables users to communicate with the chatbot in a natural and convenient way.

## II. LITERATURE SURVEY

A comprehensive review of related works was conducted to understand existing approaches to chatbot systems, campus navigation assistance, location-based services, and web-based information systems. The reviewed studies include rule-based chatbots, Natural Language Processing (NLP) techniques for user query understanding, web-based navigation systems, and map integration technologies used to provide location assistance within large institutions.

[1] Rule-Based Chatbots for Information Assistance —Many early chatbot systems used rule-based approaches where predefined questions and answers were stored in a database. When a user enters a query, the system matches keywords from the query with stored patterns and returns the corresponding response. *Gap:* Rule-based chatbots often struggle to understand complex or differently phrased queries and may fail when users ask questions in unexpected formats.

[2] NLP-Based Chatbots for Query Understanding — Recent research in chatbot development uses Natural Language Processing (NLP) techniques to better understand user input. NLP allows systems to process human language, identify key terms, and determine user intent more accurately. *Gap:* While NLP improves query understanding, many systems still require extensive training data and computational resources, which may not be practical for smaller applications.

[3] Web-Based Campus Navigation Systems —Several universities have developed web-based systems that help users navigate campus locations through interactive maps and directory services. These systems typically allow users to search for buildings, departments, or facilities and view their location on a map. Such platforms improve accessibility and help new students and visitors find their way around large campuses. *Gap:* Many of these systems rely only on search interfaces and do not provide conversational assistance, which limits user interaction and ease of use.

[4] Integration of Maps and Location-Based Services — Modern applications often integrate map services such as Google Maps to display location data visually. Map integration allows users to view routes, directions, and exact positions of buildings or landmarks. This enhances user experience by combining textual information with visual guidance. *Gap:* applications often integrate map services such as Google Maps to display location data visually. Map integration allows users to view routes, directions, and exact positions of buildings or landmarks. This enhances user experience by combining textual information with visual guidance.

From the above studies, it is evident that existing systems usually focus on either basic chat-based assistance or map-based navigation separately. However, they rarely combine both functionalities into a single interactive system. The proposed College Location Bot addresses this gap by integrating a chatbot-based query system with map navigation features. Users can ask questions about campus locations through a chat interface, and the system provides relevant responses along with a map option for visual guidance.

### III. PROPOSED SYSTEM

The proposed system is a Campus Mate – A College AI, a web-based chatbot application designed to assist students, staff, and visitors in easily locating various facilities within a college campus. Large campuses often create difficulties for new students or visitors when trying to find departments, classrooms, laboratories, administrative offices, libraries, and other important locations. The proposed system provides an automated and user-friendly solution to this problem through an interactive chat interface.

The system is developed using the Django framework, which handles the backend processing, request management, and database operations. The front-end interface is designed using HTML and CSS, providing a simple and responsive chat environment where users can interact with the bot. Users can type questions such as “Where is the library?” or “Where is the computer lab?”, and the chatbot processes the query and provides the appropriate response. The chatbot works based on keyword matching and predefined location data stored in the system.

When a user submits a query, the system analyzes the input text, identifies important keywords related to campus locations, and searches the stored data to generate the correct response. This enables the bot to provide quick and accurate information about different places inside the campus.

In addition to text responses, the proposed system includes a map navigation feature that enhances the user experience. A location icon is displayed next to the bot’s response, and when the user clicks the icon, a map page opens showing the exact location of the requested place. This visual representation helps users understand directions more clearly and improves navigation within the campus. In addition to text responses, the proposed system includes a map navigation feature that enhances the user experience.

A location icon is displayed next to the bot’s response, and when the user clicks the icon, a map page opens showing the exact location of the requested place. This visual representation helps users understand directions more clearly and improves navigation within the campus. The system also maintains a chat history during the session, allowing users to view previous questions and responses. This feature helps users keep track of the conversation and refer back to earlier information when needed.

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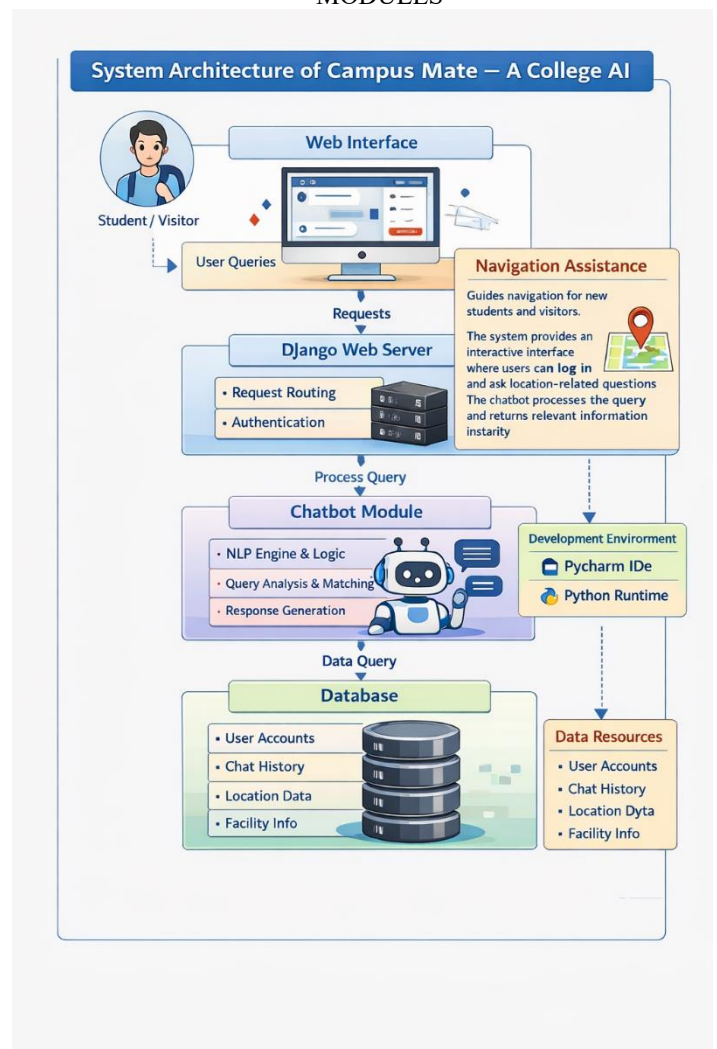
### IV. PROPOSED SOLUTION

The proposed solution is the development of a Campus Mate – A College AI, a web-based chatbot system designed to help users easily find various locations within a college campus. Many colleges have large campuses with multiple departments, laboratories, classrooms, administrative offices, and other facilities. For new students, visitors, or even staff members, locating these places can often be confusing and time-consuming. The proposed system aims to solve this problem by providing a simple and interactive platform that assists users in finding locations quickly. The system uses a chatbot interface where users can ask questions about different locations inside the college. The chatbot processes the user’s query and provides the appropriate response based on the stored location information in the system. This allows users to receive instant guidance without needing to manually search through directories or ask others for directions.

The application is developed using the Django framework, which manages the backend logic, handles user requests, and processes queries efficiently. The front-end interface is built using HTML and CSS, creating a clean and easy-to-use chat environment. Users can type their questions in the chat input box and receive responses from the bot in real time.

To improve usability, the system also includes a map navigation feature. Along with the chatbot’s response, a location icon is displayed next to the answer. When the user clicks this icon, a map page opens showing the exact location of the requested place. This helps users visually understand where the location is within the campus.

### MODULES



## V. IMPLEMENTATION

The Campus Mate – A College AI is designed using a three-tier web architecture, ensuring modular design, easy maintenance, and efficient performance. The system integrates a frontend user interface, a Django-based backend, and a structured database, forming a complete end-to-end campus navigation assistance system. The frontend provides a simple and interactive chat interface where users can type queries about different locations inside the college. It displays user messages and chatbot responses in a structured conversation format, improving usability and interaction. The interface also includes a map icon beside each bot response, which allows users to view the requested location visually. The backend, developed using the Django framework, manages the application logic, processes user queries, and retrieves the appropriate responses from stored location data. It analyzes the text entered by the user, extracts important keywords related to campus locations, and returns the relevant information. Django also handles user authentication, session management, and secure communication between components.

The database stores information about campus locations such as departments, laboratories, classrooms, and administrative offices. It ensures proper data management and quick retrieval when the chatbot processes user queries.

The system architecture ensures fast response time, reliable data storage, and easy navigation assistance, aligning with modern web-based chatbot applications. The following modules have been implemented to achieve the project objectives.

The system architecture ensures data security, real-time synchronization, and efficient reporting, aligning with modern web-based chatbot applications with extra speech to text conversion. The following modules have been implemented to achieve the project objectives:

### 1.USER INTERACTION MODULE

This module allows users to interact with the chatbot through a web-based chat interface. Users can enter questions such as “Where is the library?” or “Where is the computer lab?”. The chatbot receives the input and displays the response in the chat window. The interface is designed using HTML, CSS, and JavaScript to ensure smooth and user-friendly interaction. The chat interface also automatically scrolls to display the latest messages, maintaining a natural conversation flow.

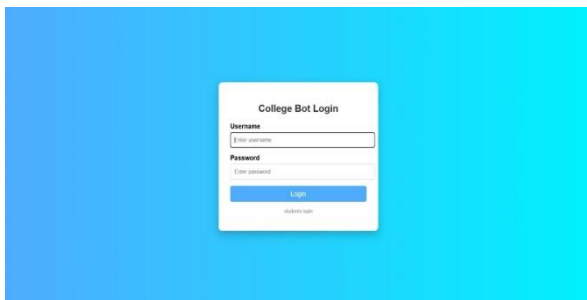


Fig 1.1 Login



Fig 1.2 Login

### 2.QUERY PROCESSING MODULE

This module processes user queries and extracts relevant keywords related to campus locations. It uses basic Natural Language Processing techniques such as text normalization and keyword matching to generate accurate responses.

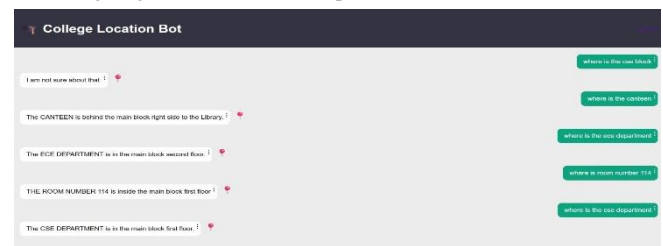


Fig 1.3 Interaction with user

### 3.FEEDBACK MANAGEMENT MODULE

This module collects and manages student feedback regarding faculty performance. It includes rating-based evaluation criteria and helps in analyzing teaching effectiveness.

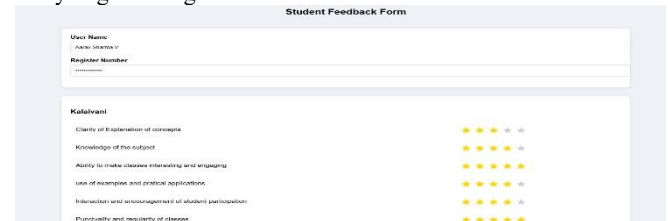


Fig 1.4 Student Feedback Form

### 4.MAP NAVIGATION MODULE

This module provides visual navigation support by integrating map functionality. It displays routes and markers for selected locations, helping users easily navigate within the campus.

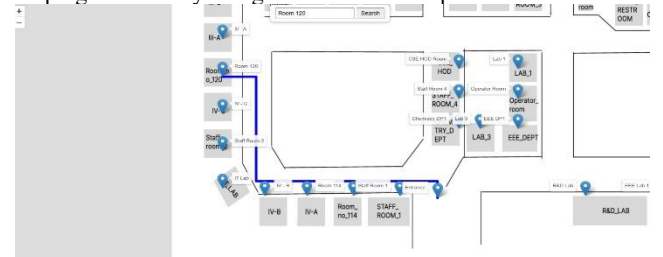


Fig 1.5 Map navigation

### 5.CHAT HISTORY MODULE

The chat history module maintains the conversation between the user and the chatbot during the session. It stores previous messages so that users can review earlier responses if needed. This improves usability by allowing users to track the conversation and avoid repeating the same questions.

### 6.STUDENT VERIFICATION MODULE

This module is used for administrative purposes to verify student records. It allows administrators to manage and update verification status efficiently through a tabular interface. This module allows administrators to manage student's verification data stored in the system. Through the Django admin panel, administrators can add, update, or remove information whenever required.

| Final CSE - No Due Verification |                  |                 |              |
|---------------------------------|------------------|-----------------|--------------|
| S.No                            | Student Name     | Register Number | Verification |
| 1                               | Aashu Sharma V   | 6176AC22UC001   | Verified ✓   |
| 2                               | Adithya Iyer S   | 6176AC22UC002   | Verified ✓   |
| 3                               | Ajyan Kumar M    | 6176AC22UC003   | Verified ✓   |
| 4                               | Ananya Reddy S   | 6176AC22UC004   | Verified ✓   |
| 5                               | Rahul Varma T    | 6176AC22UC005   | Verified ✓   |
| 6                               | Sowha Nair V     | 6176AC22UC006   | Verified ✓   |
| 7                               | Karthik Raj H    | 6176AC22UC007   | Verified ✓   |
| 8                               | Meera Krishnan P | 6176AC22UC008   | Verified ✓   |

Fig 1.6 Navigation

## 7. SOFTWARE ENVIRONMENT

### Frontend:

Designed with HTML, CSS, and JavaScript using Django templates for responsive, user-friendly chat interaction.

### Backend:

Developed in Python using the Django framework for handling application logic, query processing, and communication between components.

### Database:

**SQLite** database used for storing campus location data, chat history, and system records.

### Libraries Used:

Django, Python standard libraries, and basic text processing modules for query handling.

### IDE:

Visual Studio Code (VS Code) for code development, debugging, and testing.

## 8. CODE IMPLEMENTATION

### Step 1: Project Setup

Initialized the Django project and created an application for the chatbot. Configured the database and static files for frontend resources.

### Step 2: Database Creation

Created models for storing campus location data including department names, building names, and descriptions.

### Step 3: Chat Interface Development

Developed the chat interface using HTML and CSS, allowing users to input questions and receive responses from the chatbot.

### Step 4: Query Handling

Implemented backend logic to process user input, extract keywords, and match them with stored location data.

### Step 5: Response Generation

Generated appropriate responses based on matched keywords and displayed them in the chat interface.

### Step 6: Map Integration

Integrated Google Maps links so that users can view the exact location by clicking the map icon beside the chatbot response.

### Step 7: Chat History Management

Stored user questions and chatbot responses during the session to maintain conversation history.

### Step 8: User Authentication System

Implemented a secure user authentication system using Django's built-in authentication framework. Users are required to log in before accessing the chatbot dashboard. The system verifies user credentials, manages sessions, and provides logout functionality to ensure secure access to the chatbot services.

### Step 9: Interface Styling and Responsiveness

Enhanced the chatbot interface by applying CSS styling and responsive design techniques. The chat layout, message bubbles, and navigation buttons were optimized to provide a clean and user-friendly interface.

## 9. RESULT AND DISCUSSION

The Campus Mate – A College AI successfully provides an automated solution for navigating campus locations. The chatbot accurately responds to user queries and helps users locate departments and facilities quickly.

- Interactive chatbot interface for user queries.
- Automated response generation based on campus location data.
- Admin panel for managing location information.
- User-friendly system that improves campus accessibility.

Overall, the Campus Mate – A College AI effectively simplifies campus navigation by combining chatbot technology with location-based services. The system enhances user experience, saves time, and provides an efficient solution for students and visitors searching for locations within the college campus.

## VI. CONCLUSION

The Campus Mate – A College AI was developed to provide an efficient and user-friendly solution for navigating locations within a college campus. In large educational institutions, students and visitors often face difficulties in locating departments, classrooms, laboratories, and administrative offices. This project addresses that problem by offering an interactive chatbot system that provides quick and accurate location information. The system allows users to ask questions through a chat interface, making the process of finding locations simple and convenient. By using a web-based platform, the chatbot can be easily accessed through a browser without requiring additional software installation. The integration of chatbot technology improves communication between the user and the system by allowing natural interaction.

The project was implemented using the Django framework for backend processing and HTML and CSS for designing the user interface. The system processes user queries and retrieves relevant location information from the database. It then displays the response in a structured chat format, ensuring clarity and ease of understanding. The project was implemented using the Django framework for backend processing and HTML and CSS for designing the user interface. The system processes user queries and retrieves relevant location information from the database. It then displays the response in a structured chat format, ensuring clarity and ease of understanding.

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